GRAND-SLAM notes

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Disclaimer: These notes were made solely for the purpose of helping myself understand the GRAND-SLAM model. Relevant parts of the paper were copied or paraphrased directly. I do not seek to infringe on the authors'/journal's copyright or take credit for their work. Much of these notes were written with help from OpenAI Prism and Google Gemini. These notes assume familiarity with statistics and the EM algorithm at the level presented in Stanford CS 229 and Caltech BE/Bi 103b.

Reference: Jürges C, Dölken L, Erhard F. Dissecting newly transcribed and old RNA using GRAND-SLAM. *Bioinformatics*. 2018;34(13):i218-i226. doi:10.1093/bioinformatics/bty256

1 Overview

In the GRAND-SLAM model, each read i has:

- n_i: the number of thymine positions at which a $T \to C$ mismatch could occur,
- y_i: the number of observed $T \to C$ mismatches,
- $z_i \in \{0, 1\}$: a hidden label indicating whether the read came from old RNA ($z_i = 0$) or new RNA ($z_i = 1$).

The hierarchical model is

$$z_i \sim \text{Bernoulli}(\pi_g)$$
$$y_i \mid z_i = 0; n_i \sim \text{Binomial}(n_i, p_e), \qquad y_i \mid z_i = 1; n_i \sim \text{Binomial}(n_i, p_c)$$

where

- p_e (presumably "e" for *error*) is the average T to C mismatch rate in unlabeled RNA, and
- p_c (presumably "c" for *conversion*) is the average T to C mismatch rate in labeled RNA.

The probability mass functions are therefore

$$\begin{aligned}
P(z_i = 0 \mid p_e, p_c, \pi_g) &= P(z_i = 0 \mid \pi_g) = 1 - \pi_g, \\
P(z_i = 1 \mid p_e, p_c, \pi_g) &= P(z_i = 1 \mid \pi_g) = \pi_g, \\
P(y_i \mid z_i = 0, p_e, p_c, \pi_g; n_i) &= P(y_i \mid z_i = 0, p_e; n_i) = \text{Bin}(y_i; n_i, p_e), \\
P(y_i \mid z_i = 1, p_e, p_c, \pi_g; n_i) &= P(y_i \mid z_i = 1, p_c; n_i) = \text{Bin}(y_i; n_i, p_c)
\end{aligned}$$

where

$$\text{Bin}(y_i; n_i, p) = \binom{n_i}{y_i} p^{y_i} (1 - p)^{n_i - y_i}.$$

The joint probability mass function can therefore be decomposed as

$$\begin{aligned}
P(y_i, z_i \mid p_e, p_c, \pi_g; n_i) &= P(y_i \mid z_i, p_e, p_c, \pi_g; n_i) P(z_i \mid p_e, p_c, \pi_g; n_i) \\
&= P(y_i \mid z_i, p_e, p_c; n_i) P(z_i \mid \pi_g) \\
&= \begin{cases} (1 - \pi_g) \text{Bin}(y_i; n_i, p_e), & z_i = 0 \\ \pi_g \, \text{Bin}(y_i; n_i, p_c), & z_i = 1 \end{cases}
\end{aligned}$$

The likelihood for the observed data y_i for read i is the two-component binomial mixture (equation 1 from the paper)

$$\begin{aligned}
P(y_i \mid p_e, p_c, \pi_g; n_i) &= \sum_{z_i} P(y_i, z_i \mid p_e, p_c, \pi_g; n_i) \\
&= \sum_{z_i} P(y_i \mid z_i, p_e, p_c, \pi_g; n_i) P(z_i \mid p_e, p_c, \pi_g; n_i) \\
&= (1 - \pi_g) \, \text{Bin}(y_i; n_i, p_e) + \pi_g \, \text{Bin}(y_i; n_i, p_c).
\end{aligned}$$

The probability that a given read is nascent is simply the posterior probability

$$\begin{aligned}
P(z_i = 1 \mid y_i, p_e, p_c, \pi_g; n_i) &= \frac{P(y_i \mid z_i = 1, p_e, p_c, \pi_g; n_i) P(z_i = 1 \mid p_e, p_c, \pi_g; n_i)}{P(y_i \mid p_e, p_c, \pi_g; n_i)} \\
&= \frac{P(y_i \mid z_i = 1, p_e, p_c, \pi_g; n_i) P(z_i = 1 \mid p_e, p_c, \pi_g; n_i)}{\sum_{z_i} P(y_i \mid z_i, p_e, p_c, \pi_g; n_i) P(z_i \mid p_e, p_c, \pi_g; n_i)} \\
&= \frac{P(y_i \mid z_i = 1, p_e, p_c; n_i) P(z_i = 1 \mid \pi_g)}{\sum_{z_i} P(y_i \mid z_i, p_e, p_c, \pi_g; n_i) P(z_i \mid \pi_g)} \\
&= \frac{\pi_g \operatorname{Bin}(y_i; n_i, p_c)}{(1 - \pi_g) \operatorname{Bin}(y_i; n_i, p_e) + \pi_g \operatorname{Bin}(y_i; n_i, p_c)}
\end{aligned}$$

The primary objective is to estimate π_g, the proportion of reads from gene g that are nascent.

$$f(\pi_g \mid \mathbf{y}, p_e, p_c; \mathbf{n}) = \frac{\left(\prod_i P(y_i \mid p_e, p_c, \pi_g; n_i)\right) P(\pi_g)}{P(\mathbf{y})}$$

The GRAND-SLAM model uses a uniform prior for π_g, expressed as a beta distribution with hyperparameters $\alpha = \beta = 1$, yielding

$$f(\pi_g \mid \mathbf{y}, p_e, p_c; \mathbf{n}) = \frac{\left(\prod_i P(y_i \mid p_e, p_c, \pi_g; n_i)\right) \cdot B(\pi_g \mid \alpha = 1, \beta = 1)}{\int_0^1 \left(\prod_i P(y_i \mid p_e, p_c, \pi'_g; n_i)\right) \cdot B(\pi'_g \mid \alpha = 1, \beta = 1) d\pi'_g}$$

While the parameters π_g, p_e, and p_c can in theory be estimated together via a single EM procedure, the parameters may be poorly identifiable and the procedure may suffer from poor convergence, as described in the next section. GRAND-SLAM instead fits p_e and p_c separately, then uses fixed estimates of those parameters to estimate π_g. **Notably, GRAND-SLAM does not seem to provide any metric of variability (standard error, etc.) of its estimates of p_e and p_c.**

The posterior distribution of π_g is computed by numerically integrating the evidence (the denominator) via the trapezoidal rule, then approximated via a beta distribution with parameters α_g and β_g:

$$f(\pi_g \mid \mathbf{y}, p_e, p_c; \mathbf{n}) \approx b_g(\pi_g; \alpha_g, \beta_g) = \frac{\pi_g^{\alpha_g - 1} (1 - \pi_g)^{\beta_g - 1}}{B(\alpha_g, \beta_g)}$$

where $B(\alpha_g, \beta_g) = \frac{\Gamma(\alpha_g)\Gamma(\beta_g)}{\Gamma(\alpha_g + \beta_g)}$ is the beta function.

2 Parameter Estimation

2.1 Standard EM procedure

Step 1: Initialize the parameters.

Choose initial guesses $p_e^{(0)}$, $p_c^{(0)}$, and $\pi_g^{(0)}$ for every gene g.

Step 2: E-step.

For each read i from gene g, compute the posterior probability that it came from the *new* RNA component:

$$\begin{aligned} r_i^{(t)} &= P(z_i = 1 \mid y_i, p_e^{(t)}, p_c^{(t)}, \pi_g^{(t)}; n_i) \\ &= \frac{\pi_g^{(t)} \operatorname{Bin}(y_i; n_i, p_c^{(t)})}{(1-\pi_g^{(t)}) \operatorname{Bin}(y_i; n_i, p_e^{(t)}) + \pi_g^{(t)} \operatorname{Bin}(y_i; n_i, p_c^{(t)})} \\ &= \frac{\pi_g^{(t)} \binom{n_i}{y_i} \left(p_c^{(t)}\right)^{y_i} \left(1-p_c^{(t)}\right)^{n_i-y_i}}{(1-\pi_g^{(t)}) \binom{n_i}{y_i} \left(p_e^{(t)}\right)^{y_i} \left(1-p_e^{(t)}\right)^{n_i-y_i} + \pi_g^{(t)} \binom{n_i}{y_i} \left(p_c^{(t)}\right)^{y_i} \left(1-p_c^{(t)}\right)^{n_i-y_i}} \\ &= \frac{\pi_g^{(t)} \left(p_c^{(t)}\right)^{y_i} \left(1-p_c^{(t)}\right)^{n_i-y_i}}{(1-\pi_g^{(t)}) \left(p_e^{(t)}\right)^{y_i} \left(1-p_e^{(t)}\right)^{n_i-y_i} + \pi_g^{(t)} \left(p_c^{(t)}\right)^{y_i} \left(1-p_c^{(t)}\right)^{n_i-y_i}}. \end{aligned}$$

Then $1 - r_i^{(t)}$ is the posterior probability that the read came from the *old* RNA component.

Intuitively, this step "softly assigns" each read to one of the two components.

Step 3: M-step for the mixing proportions.

For each gene g, update the estimated fraction of new RNA by averaging those soft assignments over all reads from that gene:

$$\pi_g^{(t+1)} = \frac{1}{m_g} \sum_{i \in g} r_i^{(t)},$$

where m_g is the number of reads mapped to gene g.

Step 4: M-step for p_e.

Update the mismatch probability in the old-RNA component by the weighted fraction of mismatches assigned to that component:

$$p_e^{(t+1)} = \frac{\sum_i (1 - r_i^{(t)}) y_i}{\sum_i (1 - r_i^{(t)}) n_i}.$$

Step 5: M-step for p_c.

Update the mismatch probability in the new-RNA component analogously:

$$p_c^{(t+1)} = \frac{\sum_i r_i^{(t)} y_i}{\sum_i r_i^{(t)} n_i}.$$

Step 6: Iterate until convergence.

Repeat the E-step and M-step until the parameter updates are tiny or the log-likelihood stops changing.

This is exactly the usual EM recipe for a mixture model: estimate hidden memberships, then re-estimate the component parameters from those memberships.

2.1.1 Why the standard EM approach fails here

Under standard 4sU-labeling conditions, the authors caution that "$y = 0$ [observing no T to C mismatches] is the mode of both component distributions of the binomial mixture model" because both p_e and p_c are small. Therefore, the dataset is dominated by observations that are simultaneously very plausible under the old-RNA model and the new-RNA model. For such reads, the E-step responsibility is

$$r_i^{(t)} = \frac{\pi_g^{(t)}(1-p_c^{(t)})^{n_i}}{(1-\pi_g^{(t)})(1-p_e^{(t)})^{n_i} + \pi_g^{(t)}(1-p_c^{(t)})^{n_i}},$$

which does not strongly favor either component when both $p_e^{(t)}$ and $p_c^{(t)}$ are small. The likelihood surface therefore has a long, flat ridge rather than a sharp optimum, and EM makes very little progress because its soft assignments barely change from one iteration to the next. This is the practical meaning of the paper's claim that "standard approaches to estimate pe and pc based on the expectation maximization (EM) algorithm failed."

GRAND-SLAM's workaround is to first estimate p_e separately, then look only at the tail with sufficiently many mismatches, where the old-RNA component contributes negligibly. In that tail, the data are much more informative about p_c, so the estimation problem becomes identifiable enough to solve.

2.2 Estimating p_e

Methods proposed by Jürges et al.:

- Using an additional sample without 4sU labeling, such as a timepoint 0 sample or spike-in RNAs.
- Linear regression model to predict T to C error rate from the other error rates (e.g., T to A and T to G error rates).

2.3 Estimating p_c

2.3.1 Data preprocessing

The sufficient statistics for parameter estimation are collected in a matrix $A^{(g)}$. Each entry $A^{(g)}_{k,n}$ is the number of reads mapped to a genomic region within gene g containing n thymines with k observed T to C mismatches. Aggregating counts across genes generates the matrix

$$A = \{A_{k,n}\} = \sum_g A^{(g)}.$$

Consider reads with a given number of thymine sites n (i.e., a column of A). Let $e_{k,n}$ be the number of reads with k mismatches expected to originate from unlabeled RNA. Since $p_e < p_c$ (the underlying assumption of 4sU labeling), there exists a threshold number of T to C mismatches k at and above which only a minor fraction of reads (say, 1%) originates from the unlabeled component (i.e., $e_{k,n} \leq 0.01 a_{k,n}$). Based on this threshold, we partition the cells (k, n) into 2 mutually exclusive sets:

- $R = \{(k, n) : e_{k,n} \leq 0.01 a_{k,n}\}$: the inclusion set where unlabeled RNAs are expected to contribute a minor fraction of reads
  - Let $R_n = \{k : (k, n) \in R\}$ be the retained mismatch counts for reads with n possible conversion sites.
- $X = \{(k, n) : e_{k,n} > 0.01\, a_{k,n}\}$: the exclusion set where unlabeled RNAs are expected to contribute more than 1% of reads
  - Let $X_n = \{k : (k, n) \in X\}$ be the excluded mismatch counts for reads with n possible conversion sites.

How is $e_{k,n}$ defined? Let $M_n = \sum_{k'} A_{k',n}$ be the column sum of A: the total number of reads (across all genes) containing n thymine sites. If all of these reads were generated from the unlabeled RNA component ($z_i = 0$), their counts could be collected into a matrix $E = \{E_{k,n}\}$, with the same column sums M_n as in the matrix A, but a different distribution within each column. Now, consider a particular value of n. Given p_e, whether each of M_n reads is observed with k mismatches (and is added to bin $E_{k,n}$) is an independent Bernoulli random variable

$$\mathbf{1}\{y_i = k\} \sim \text{Bernoulli}(P(y_i = k \mid z_i = 0; n) = \text{Bin}(k; n, p_e))$$

Therefore, the count of each bin is

$$E_{k,n} = \sum_{i \,|\, n_i = n} \mathbf{1}\{y_i = k\}$$

with expected value

$$\begin{aligned} e_{k,n} = \mathbb{E}[E_{k,n}] &= \mathbb{E}\left[\sum_{i \mid n_i = n} \mathbf{1}\{y_i = k\}\right] \\ &= \sum_{i \mid n_i = n} \mathbb{E}[\mathbf{1}\{y_i = k\}] \\ &= \sum_{i \mid n_i = n} \mathrm{Bin}(k; n, p_e) \\ &= M_n \,\mathrm{Bin}(k; n, p_e) \\ &= \mathrm{Bin}(k; n, p_e) \sum_{k'} a_{k',n}. \end{aligned}$$

2.3.2 Model: Observed data, latent variables

Let $A^{(\text{unlabeled})}$ and A^* be the unlabeled and labeled counts such that $A^{(\text{unlabeled})} + A^* = A$. ($A^{(\text{unlabeled})}$ is only introduced here once as a placeholder; it is not used further in the model.)

Consider a latent variable model of the labeled counts A^*, where the entries of A^* are partitioned into unobserved and observed sets

- $A^*_{\text{unobs}} = \{a^*_{k,n} \mid (k,n) \in X\}$. For fixed n, denote the corresponding subset as $A^*_{\text{unobs},n}$.
- $A^*_{\text{obs}} = \{a^*_{k,n} \mid (k,n) \in R\} = \{a_{k,n} \mid (k,n) \in R\}$

such that $A^*_{\text{unobs}} \cup A^*_{\text{obs}} = A^*$ (here we treat $A^* = \{a^*_{k,n}\}$ as a set rather than a matrix). GRAND-SLAM treats the retained cell counts (defined in the section above) as if they are all generated from the 4sU-labeled component, so the retained cell counts correspond to the observed labeled counts A^*_{obs}, hence $a^*_{k,n} = a_{k,n}$ for $(k,n) \in R$.

As an example, consider a fixed n (i.e., a column of A), say $n = 3$ possible conversion sites. Let's say that the threshold value of k was 2, such that reads with 2 or 3 conversions out of 3 possible conversion sites are now assumed to be from the 4sU-labeled component. (We assume that reads with 0 or 1 conversions may have come from both the unlabeled and 4sU-labeled components.) Therefore, we assume that $A^*_{2,3} = A_{2,3}$ and $A^*_{3,3} = A_{3,3}$ – these are the observed variables. $A^*_{0,3}$ and $A^*_{1,3}$ are the unobserved latent variables.

2.3.3 Complete-data likelihood

If the complete counts $a^*_{k,n}$ were known, estimating p_c would be easy. The complete-data likelihood is

$$P(A^* \mid p_c) = \prod_n \left[\frac{N_n!}{\prod_{k=0}^n a^*_{k,n}!} \prod_{k=0}^n \mathrm{Bin}(k; n, p_c)^{a^*_{k,n}}\right].$$

so the complete-data log-likelihood is

$$\begin{aligned}
\ell_c(p_c) &= \sum_n \sum_{k=0}^{n} a^*_{k,n} \log \mathrm{Bin}(k; n, p_c) + \text{const} \\
&= \sum_n \sum_{k=0}^{n} a^*_{k,n} \left(\log \binom{n}{k} + k \log p_c + (n-k) \log(1-p_c) \right) + \text{const} \\
&= \sum_n \sum_{k=0}^{n} a^*_{k,n} \left(k \log p_c + (n-k) \log(1-p_c) \right) + \text{const} \\
&= \log p_c \left(\sum_n \sum_{k=0}^{n} a^*_{k,n} k \right) + \log(1-p_c) \left(\sum_n \sum_{k=0}^{n} a^*_{k,n} (n-k) \right) + \text{const}
\end{aligned}$$

Setting the first derivative of the log-likelihood equal to zero, we get

$$0 = \frac{\partial \ell_c}{\partial p_c} = \frac{1}{p_c} \left(\sum_n \sum_{k=0}^{n} a^*_{k,n} k \right) - \frac{1}{1-p_c} \left(\sum_n \sum_{k=0}^{n} a^*_{k,n} (n-k) \right).$$

Rearranging,

$$\frac{1-p_c}{p_c} = \frac{\sum_n \sum_{k=0}^{n} a^*_{k,n} (n-k)}{\sum_n \sum_{k=0}^{n} a^*_{k,n} k},$$

so

$$p_c = \frac{\sum_n \sum_{k=0}^{n} a^*_{k,n} k}{\sum_n \sum_{k=0}^{n} a^*_{k,n} k + \sum_n \sum_{k=0}^{n} a^*_{k,n} (n-k)}.$$

Since

$$\sum_n \sum_{k=0}^{n} a^*_{k,n} k + \sum_n \sum_{k=0}^{n} a^*_{k,n} (n-k) = \sum_n \sum_{k=0}^{n} a^*_{k,n} n,$$

this simplifies to

$$\widehat{p}_c = \frac{\sum_{k,n} k \, a^*_{k,n}}{\sum_{k,n} n \, a^*_{k,n}}.$$

So with complete data, the solution is just "total conversions divided by total conversion opportunities."

2.3.4 Observed-data likelihood

Because the excluded cells are unobserved, the actual observed-data likelihood is obtained by summing over all possible unobserved low-mismatch counts. For each fixed n, introduce:

- $O_n = \sum_{k \in R_n} a_{k,n}$, the total number of retained reads with n possible conversion sites;
- $q_n(p_c) = \sum_{k \in R_n} \mathrm{Bin}(k; n, p_c)$, the probability that a read from a 4sU-labeled RNA contains enough mismatches and is retained (observed).

Then the retained counts follow a truncated multinomial model

$$P(A^*_{\text{obs}} \mid p_c) = \prod_n \left[\frac{O_n!}{\prod_{k \in R_n} a_{k,n}!} \prod_{k \in R_n} \left(\frac{\text{Bin}(k; n, p_c)}{q_n(p_c)} \right)^{a_{k,n}} \right]$$

and

$$\ell_{\text{obs}}(p_c) = \sum_n \sum_{k \in R_n} a_{k,n} \log \text{Bin}(k; n, p_c) - \sum_n O_n \log q_n(p_c) + \text{const.}$$

Compared to the complete-data likelihood, this observed-data likelihood is harder to optimize due to the normalizing terms $q_n(p_c)$. EM avoids this difficulty by repeatedly reconstructing the unobserved counts from the current parameter value and then maximizing the simpler complete-data criterion.

We first outline the application of the EM algorithm to this latent variable model, starting with the observed-data log likelihood

$$\begin{aligned}
\ell_{\text{obs}}(p_c; A^*_{\text{obs}}) &= \log p(A^*_{\text{obs}} \mid p_c) \\
&= \log \sum_{A^*_{\text{unobs}}} p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c) \\
&= \log \sum_{A^*_{\text{unobs}}} Q(A^*_{\text{unobs}}) \frac{p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c)}{Q(A^*_{\text{unobs}})} \\
&\geq \sum_{A^*_{\text{unobs}}} Q(A^*_{\text{unobs}}) \log \frac{p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c)}{Q(A^*_{\text{unobs}})} \\
&= \text{ELBO}(A^*_{\text{obs}}, Q, p_c).
\end{aligned}$$

To make the inequality tight, we set the proposal distribution to the posterior distribution of the unobserved data

$$Q(A^*_{\text{unobs}}) = Q(A^*_{\text{unobs}} \mid A^*_{\text{obs}}, p_c^{(t)}) = p(A^*_{\text{unobs}} \mid A^*_{\text{obs}}, p_c^{(t)}).$$

The E-step defines the expected value function

$$\begin{aligned}
\mathbb{Q}(p_c \mid p_c^{(t)}) &= \mathbb{E}_{A^*_{\text{unobs}} \sim Q} \log p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c) \\
&= \sum_{A^*_{\text{unobs}}} Q(A^*_{\text{unobs}} \mid A^*_{\text{obs}}, p_c^{(t)}) \log p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c)
\end{aligned}$$

The M-step then maximizes the expected value function with respect to the parameter p_c

$$p_c^{(t+1)} = \arg\max_{p_c} \mathbb{Q}(p_c \mid p_c^{(t)}).$$

To apply the EM algorithm, we need expressions for the joint log likelihood $\log p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c)$ and the latent variable posterior $p(A^*_{\text{unobs}} \mid A^*_{\text{obs}}, p_c^{(t)})$. The expression for the joint log likelihood

was already presented earlier in the "Complete-data likelihood" section:

$$\begin{aligned}\log p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c) &= \sum_n \sum_{k=0}^{n} a^*_{k,n} \log \text{Bin}(k; n, p_c) + \text{const} \\ &= \sum_n \left(\sum_{k \in R_n} a_{k,n} \log \text{Bin}(k; n, p_c) + \sum_{k \in X_n} a^*_{k,n} \log \text{Bin}(k; n, p_c) \right) + \text{const}\end{aligned}$$

The GRAND-SLAM paper does not specify the specific likelihood model for unobserved counts A^*_{unobs}; it only presents the E-step and M-step updates. Here, I present one likelihood model suggested by Gemini that is consistent with the GRAND-SLAM E- and M-step updates; other consistent models may exist. To express the posterior distribution for A^*_{unobs} explicitly, consider reads from the labeled RNA component with a fixed number of conversion opportunities n. Each read is a Bernoulli trial with a probability of "success" (retained and observed) $q_n(p_c)$. The number of "failures" (reads excluded and unobserved) $U_n = \sum_{k \in X_n} a^*_{k,n}$ before observing O_n "successes" therefore follows a Negative Binomial distribution. Given U_n, the joint distribution of the specific unobserved counts $\{a^*_{k,n}\}_{k \in X_n}$ scattered across the exclusion bins is a Multinomial distribution with probabilities proportional to $\text{Bin}(k; n, p_c)$. Therefore, for each n, $A^*_{\text{unobs},n}$ is drawn from a hierarchical model

$$\begin{aligned} U_n \mid A^*_{\text{obs}}, p_c &\sim \text{NB}(O_n, q_n(p_c)) \\ A^*_{\text{unobs},n} \mid U_n &\sim \text{Multinomial}\left(U_n, \left\{ \frac{\text{Bin}(k; n, p_c)}{1 - q_n(p_c)} \right\}_{k \in X_n} \right)\end{aligned}$$

(More explicitly, $A^*_{\text{unobs},n} = (a^*_{0,n}, \ldots, a^*_{k^*,n})$ where $k^* = \max_k\{k \in X_n\}$.) Note that for a given set of unobserved counts A^*_{unobs}, $U_n = \sum_{k \in X_n} a^*_{k,n}$ is fixed. Therefore, the marginal probability of A^*_{unobs} is simply the joint probability evaluated exactly at U_n:

$$\begin{aligned} p(A^*_{\text{unobs}} \mid A^*_{\text{obs}}, p_c) &= \prod_n p(A^*_{\text{unobs},n} \mid A^*_{\text{obs}}, p_c) \\ &= \prod_n p(A^*_{\text{unobs},n}, U_n \mid A^*_{\text{obs}}, p_c) \\ &= \prod_n p(A^*_{\text{unobs},n} \mid U_n, A^*_{\text{obs}}, p_c) p(U_n \mid A^*_{\text{obs}}, p_c) \\ &= \prod_n \left[\text{Multinomial}\left(A^*_{\text{unobs},n}; U_n, \left\{ \frac{\text{Bin}(k; n, p_c)}{1 - q_n(p_c)} \right\}_{k \in X_n} \right) \text{NB}(U_n; O_n, q_n(p_c)) \right]\end{aligned}$$

(This expression can be fully expanded[1], but it is never used directly.) We are now ready to define

[1]

$$\begin{aligned} p(A^*_{\text{unobs}} \mid A^*_{\text{obs}}, p_c) &= \prod_n \left[\frac{U_n!}{\prod_{k \in X_n} a^*_{k,n}!} \left(\prod_{k \in X_n} \left(\frac{\text{Bin}(k; n, p_c)}{1 - q_n(p_c)} \right)^{a^*_{k,n}} \right) \binom{O_n + U_n - 1}{U_n} (q_n(p_c))^{O_n} (1 - q_n(p_c))^{U_n} \right] \\ &= \prod_n \left[\frac{U_n!}{\prod_{k \in X_n} a^*_{k,n}!} \left(\prod_{k \in X_n} (\text{Bin}(k; n, p_c))^{a^*_{k,n}} \right) \frac{(O_n + U_n - 1)!}{U_n!(O_n - 1)!} (q_n(p_c))^{O_n} \right] \\ &= \prod_n \left[\frac{(O_n + U_n - 1)!}{(O_n - 1)! \prod_{k \in X_n} a^*_{k,n}!} (q_n(p_c))^{O_n} \prod_{k \in X_n} (\text{Bin}(k; n, p_c))^{a^*_{k,n}} \right]\end{aligned}$$

the expected value function:

$$\begin{aligned}
\mathbb{Q}(p_c \mid p_c^{(t)}) &= \mathbb{E}_{A^*_{\text{unobs}} \sim Q} \log p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c) \\
&= \mathbb{E}_{A^*_{\text{unobs}} \sim Q} \left[\sum_n \left(\sum_{k \in R_n} a_{k,n} \log \text{Bin}(k; n, p_c) + \sum_{k \in X_n} a^*_{k,n} \log \text{Bin}(k; n, p_c) \right) + \text{const} \right] \\
&= \sum_n \left(\sum_{k \in R_n} a_{k,n} \log \text{Bin}(k; n, p_c) + \sum_{k \in X_n} \mathbb{E}_{A^*_{\text{unobs}} \sim Q}[a^*_{k,n}] \log \text{Bin}(k; n, p_c) \right) + \text{const}
\end{aligned}$$

where the last equality uses the linearity of expectation. This equation shares the exact algebraic form of the complete-data log-likelihood, but with the cell counts replaced by their expected values under Q. This naturally defines the expected counts $a^{(t+1)}_{k,n}$:

$$a^{(t+1)}_{k,n} = \begin{cases} a_{k,n} & k \in R_n \\ \mathbb{E}_Q[a^*_{k,n}] & k \in X_n \end{cases}$$

To evaluate $\mathbb{E}_Q[a^*_{k,n}]$ for the unobserved bins, use the law of total expectation[2] conditioning on the intermediate variable U_n (the total number of unobserved reads for a given n):

$$\mathbb{E}_Q[a^*_{k,n}] = \mathbb{E}_{U_n \sim Q}\big[\mathbb{E}[a^*_{k,n} \mid U_n]\big]$$

The inner conditional expectation is the mean of a single component in a Multinomial distribution, which is the total number of trials U_n multiplied by the probability of falling into category k:

$$\mathbb{E}[a^*_{k,n} \mid U_n] = U_n \frac{\text{Bin}(k; n, p_c^{(t)})}{1 - q_n(p_c^{(t)})}$$

Substitute the inner expectation back into the outer expectation. Because U_n is the only random variable, factor out the probabilities:

$$\mathbb{E}_Q[a^*_{k,n}] = \mathbb{E}_{U_n}[U_n] \left(\frac{\text{Bin}(k; n, p_c^{(t)})}{1 - q_n(p_c^{(t)})} \right)$$

The expectation $\mathbb{E}_{U_n}[U_n]$ is the mean of a Negative Binomial distribution representing the number of failures before O_n successes, with a success probability of $q_n(p_c^{(t)})$. The mean of this distribution is:

$$\mathbb{E}_{U_n}[U_n] = O_n \frac{1 - q_n(p_c^{(t)})}{q_n(p_c^{(t)})}$$

Thus, we have

$$\begin{aligned}
\mathbb{E}_Q[a^*_{k,n}] &= \left(O_n \frac{1 - q_n(p_c^{(t)})}{q_n(p_c^{(t)})} \right) \left(\frac{\text{Bin}(k; n, p_c^{(t)})}{1 - q_n(p_c^{(t)})} \right) \\
&= \frac{O_n \text{Bin}(k; n, p_c^{(t)})}{q_n(p_c^{(t)})} \\
&= \frac{\left(\sum_{k' \in R_n} a_{k',n}\right) \text{Bin}(k; n, p_c^{(t)})}{\sum_{k' \in R_n} \text{Bin}(k'; n, p_c^{(t)})}
\end{aligned}$$

[2]Law of total expectation: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$

This finally gives equation (5) from the GRAND-SLAM paper,

$$a_{k,n}^{(t+1)} = \begin{cases} a_{k,n} & k \in R_n \\ \frac{(\sum_{k' \in R_n} a_{k',n}) \mathrm{Bin}(k; n, p_c^{(t)})}{\sum_{k' \in R_n} \mathrm{Bin}(k'; n, p_c^{(t)})} & k \in X_n \end{cases}$$

which we plug into the expected value function

$$\begin{aligned} \mathbb{Q}(p_c \mid p_c^{(t)}) &= \mathbb{E}_{A^*_{\text{unobs}} \sim Q} \log p(A^*_{\text{obs}}, A^*_{\text{unobs}} \mid p_c) \\ &= \sum_n \sum_{k=0}^{n} a_{k,n}^{(t+1)} \log \mathrm{Bin}(k; n, p_c) + \text{const.} \end{aligned}$$

In the M-step, we maximize $\mathbb{Q}(p_c \mid p_c^{(t)})$ over p_c. Since the form of $\mathbb{Q}(p_c \mid p_c^{(t)})$ is identical to the form of the complete-data log-likelihood $\ell_c(p_c)$, its maximum is of the same form as that of the complete-data log-likelihood

$$p_c^{(t+1)} = \frac{\sum_{k,n} k\, a_{k,n}^{(t+1)}}{\sum_{k,n} n\, a_{k,n}^{(t+1)}}.$$

2.3.5 GRAND-SLAM's bisection scheme

The GRAND-SLAM paper writes, "We noticed that running the EM algorithm led to extremely slow convergence rates. Thus, we use the following bisection scheme instead: For the search interval $[l, r]$ (starting with $l = 0$ and $r = 1$), we set $p^{(t)} = \frac{l+r}{2}$ and compute $p^{(t+1)}$ by a single EM iteration. If $p^{(t+1)} < p^{(t)}$, we continue with the search interval $[l, p^{(t)}]$, otherwise with $[p^{(t)}, r]$. We stop if $r - l < 10^{-8}$."

Properties of this bisection scheme:

- Guaranteed to converge within an interval of 10^{-8} in exactly $\lceil \log_2(10^8) \rceil = 27$ iterations.
- Does not share the monotonically-improving likelihood guarantee of the standard EM algorithm. (need proof) However, since the log-likelihood function is concave and unimodal, the bisection scheme is still guaranteed to converge to the global maximum $\hat{p}$.
- Does not assume that $p^{(t+1)}$ falls in the interval $[l, r]$; the next sub-interval is chosen solely based on the boolean condition $p^{(t+1)} < p^{(t)}$. (need proof) Nonetheless, for the GRAND-SLAM model, $p^{(t+1)}$ will always fall inside $[l, r]$, since for a concave log-likelihood of a scalar parameter, a single EM step will never overshoot the maximum likelihood estimate.

3 Estimating RNA production and decay

State variable

- a: total abundance of RNA species (dimensions = reads)

Parameters

- σ: transcription rate (dimensions = reads / time)
- δ: decay rate (dimensions = 1 / time)

The governing first-order ODE model is

$$\frac{da}{dt} = \sigma - \delta a(t), \qquad a(0) = a_0$$

with solution to the initial value problem

$$a(t) = \left(a_0 - \frac{\sigma}{\delta}\right) e^{-t\delta} + \frac{\sigma}{\delta}$$

and steady-state $\frac{\sigma}{\delta}$. This ODE model is used to separately describe the abundance of pre-existing RNA a_{pre} and newly synthesized RNA a_{new}.

- Pre-existing RNA is assumed to start with steady-state abundance and then decay without a production term. **This corresponds to the physical assumption that all new RNAs synthesized after the start of 4sU pulse are in the 4sU-labeled component with T to C conversion rate p_c.**
$$a_{\text{pre}}(t) = \frac{\sigma}{\delta} e^{-t\delta}$$
- New RNA starts from an initial abundance of 0.
$$a_{\text{new}}(t) = -\frac{\sigma}{\delta} e^{-t\delta} + \frac{\sigma}{\delta}$$

The total abundance is maintained at steady-state, $a_{\text{pre}}(t) + a_{\text{new}}(t) = \frac{\sigma}{\delta}$. At any time t, the proportion of new RNA out of total RNA is

$$\begin{aligned}
\pi(t) &= \frac{a_{\text{new}}(t)}{a_{\text{new}}(t) + a_{\text{pre}}(t)} \\
&= \frac{-\frac{\sigma}{\delta} e^{-t\delta} + \frac{\sigma}{\delta}}{-\frac{\sigma}{\delta} e^{-t\delta} + \frac{\sigma}{\delta} + \frac{\sigma}{\delta} e^{-t\delta}} \\
&= 1 - e^{-t\delta}
\end{aligned}$$

3.1 Decay rate

This can be used to estimate the decay rate (here, expressing the decay rate δ as a function of the random variable π, for which we have approximated its posterior distribution with a beta distribution $b_g(\pi_g; \alpha_g, \beta_g)$)

$$\delta_t(\pi) = -\frac{1}{t} \log(1 - \pi).$$

Since $\delta_t(\pi)$ is a differentiable and invertible function (with inverse $\pi_t(\delta) = 1 - e^{-t\delta}$), the posterior[3] density function of δ_t is given by the change of variables formula (also known as the method of transformations)

$$\begin{aligned}
d_g(\delta_g; \alpha_g, \beta_g) &= \beta_g(\pi_t(\delta_g); \alpha_g, \beta_g) \left| \frac{d\pi_t(\delta_g)}{d\delta_g} \right| \\
&= \frac{(\pi_t(\delta_g))^{\alpha_g - 1}(1 - (\pi_t(\delta_g)))^{\beta_g - 1}}{B(\alpha_g, \beta_g)} t e^{-t\delta_g} \\
&= \frac{(1 - e^{-t\delta_g})^{\alpha_g - 1}(e^{-t\delta_g})^{\beta_g - 1}}{B(\alpha_g, \beta_g)} t e^{-t\delta_g} \\
&= \frac{t}{B(\alpha_g, \beta_g)} (1 - e^{-t\delta_g})^{\alpha_g - 1} e^{-t\beta_g\delta_g + t\delta_g} e^{-t\delta_g} \\
&= \frac{t}{B(\alpha_g, \beta_g)} (1 - e^{-t\delta_g})^{\alpha_g - 1} e^{-t\beta_g\delta_g}
\end{aligned}$$

with log posterior density

$$\ell(\delta_g) = (\alpha_g - 1)\log(1 - e^{-t\delta_g}) - t\beta_g\delta_g + \text{const.}$$

Generalizing to multiple observations across different timepoints is a bit tricky. Since the (approximate) posterior of $\pi_g^{(i)}$ at timepoint t_i is a $\text{Beta}(\alpha_g^{(i)}, \beta_g^{(i)})$ distribution, and a uniform prior $p(\pi_g^{(i)}) \sim \text{Beta}(1, 1)$ was used, the (approximate) likelihood is

$$\begin{aligned}
L(\pi_g^{(i)} \mid \mathbf{y}^{(i)}, p_e^{(i)}, p_c^{(i)}) &\propto \frac{f(\pi_g^{(i)} \mid \mathbf{y}^{(i)}, p_e^{(i)}, p_c^{(i)})}{p(\pi_g^{(i)})} \\
&\propto f(\pi_g^{(i)} \mid \mathbf{y}^{(i)}, p_e^{(i)}, p_c^{(i)}) \\
&\propto \left(\pi_g^{(i)}\right)^{\alpha_g^{(i)} - 1} (1 - \pi_g^{(i)})^{\beta_g^{(i)} - 1}.
\end{aligned}$$

Because likelihoods are functions of the *data* rather than the parameters, they do not require a Jacobian determinant when changing variables.[4] We therefore substitute the relationship $\pi_g^{(i)} = 1 - e^{-t_i\delta_g^{(i)}}$ directly into the likelihood:

$$\begin{aligned}
L(\delta_g^{(i)} | \mathbf{y}^{(i)}, p_e^{(i)}, p_c^{(i)}) &\propto (1 - e^{-t_i\delta_g^{(i)}})^{\alpha_g^{(i)} - 1}(1 - (1 - e^{-t_i\delta_g^{(i)}}))^{\beta_g^{(i)} - 1} \\
&= (1 - e^{-t_i\delta_g^{(i)}})^{\alpha_g^{(i)} - 1}(e^{-t_i\delta_g^{(i)}})^{\beta_g^{(i)} - 1} \\
&= (1 - e^{-t_i\delta_g^{(i)}})^{\alpha_g^{(i)} - 1} e^{-t_i(\beta_g^{(i)} - 1)\delta_g^{(i)}}
\end{aligned}$$

The joint likelihood over all samples is the product of the individual likelihoods:

$$L_{\text{joint}}(\delta_g | \mathbf{y}, p_e, p_c) = \prod_{i=1}^{m} L(\delta_g | \mathbf{y}^{(i)}, p_e^{(i)}, p_c^{(i)})$$

$$l_{\text{joint}}(\delta_g) = \log L_{\text{joint}}(\delta_g | \mathbf{y}, p_e, p_c) = \sum_{i=1}^{m} \left(\alpha_g^{(i)} - 1\right) \log\left(1 - e^{-t_i\delta_g}\right) - t_i(\beta_g^{(i)} - 1)\delta_g$$

[3] $d(\delta; \alpha_g, \beta_g)$ is a *posterior* density because it is a transformation of the *posterior* density of π_g

[4] See, for example, `https://stats.stackexchange.com/a/174979` or sections 2.8-2.9 of *In All Likelihood* (2001) by Yudi Pawitan.

The posterior distribution for δ_g is proportional to this likelihood multiplied by a *single* prior $P(\delta_g)$

$$P(\delta_g \mid \mathbf{y}, p_e, p_c) \propto P(\delta_g) \cdot L_{\text{joint}}(\delta_g | \mathbf{y}, p_e, p_c).$$

and the log-posterior is

$$\begin{aligned}\log P(\delta_g \mid \mathbf{y}, p_e, p_c) &= \log\left(P(\delta_g)\right) + l_{\text{joint}}(\delta_g) + \text{const} \\ &= \log\left(P(\delta_g)\right) + \sum_{i=1}^{m} \left(\alpha_g^{(i)} - 1\right) \log\left(1 - e^{-t_i \delta_g}\right) - t_i(\beta_g^{(i)} - 1)\delta_g + \text{const}\end{aligned}$$

The MAP estimate of δ_g is then $\arg\max_{\delta_g} \log P(\delta_g \mid \mathbf{y}, p_e, p_c)$, to be found numerically.

Compare this derivation of the log-posterior density with that suggested by Jürges et al. equation (17):

$$\ell_{\text{agg}}(\delta_g) = \sum_{i=1}^{m} (\alpha_g^{(i)} - 1) \log\left(1 - e^{-t_i \delta_g}\right) - t_i \beta_g^{(i)} \delta_g + \text{const.}$$

By splitting the $-t_i \beta_g^{(i)} \delta_g$ term into $-t_i(\beta_g^{(i)} - 1)\delta_g - t_i \delta_g$, we can re-write the paper's equation as

$$\begin{aligned}\ell_{\text{agg}}(\delta_g) &= \sum_{i=1}^{m} (\alpha_g^{(i)} - 1) \log\left(1 - e^{-t_i \delta_g}\right) - t_i(\beta_g^{(i)} - 1)\delta_g - t_i \delta_g + \text{const} \\ &= l_{\text{joint}}(\delta_g) - \delta_g \sum_{i=1}^{m} t_i + \text{const.}\end{aligned}$$

Comparing this to the true log-posterior derived above reveals the implied prior for the Jürges et al. posterior to be

$$\log P(\delta_g; \{t_i\}_{i=1}^m) = -\delta_g \sum_{i=1}^{m} t_i \quad (+\text{const})$$
$$P(\delta_g; \{t_i\}_{i=1}^m) \propto e^{-\delta_g \sum_{i=1}^m t_i}.$$

This means that the more timepoints sampled, or the later the timepoints sampled, the more aggressively the model's implied prior forces the estimate of δ_g toward 0 (infinitely long half-lives). **In short, the timepoint-aggregated estimator $\hat{\delta}_g$ proposed by the authors is not true a MAP estimate and is biased by the timepoints chosen entirely independent of the underlying biological data.**

Another way to see this is to exponentiate the log-posterior proposed by the paper to obtain the proposed posterior density

$$d_g^{(\text{agg})}(\delta_g; \alpha_g^{(1)}, \beta_g^{(1)}, ..., \alpha_g^{(m)}, \beta_g^{(m)}) = \prod_{i=1}^{m} d_g(\delta_g; \alpha_g^{(i)}, \beta_g^{(i)}).$$

Since each term $d_g(\delta; \alpha_g^{(i)}, \beta_g^{(i)})$ is already a posterior density, this product erroneously applies the $P(\delta_g)$ prior m times. Furthermore, although the likelihood is invariant under reparameterization, the prior is not; applying the change of variables to the uniform prior on $\pi_g^{(i)}$ yields an exponential prior on $\delta_g^{(i)}$

$$P(\delta_g^{(i)}) = \underbrace{1}_{\text{Uniform}} \cdot \left| \frac{d\pi_g^{(i)}(\delta_g^{(i)})}{d\delta_g^{(i)}} \right| = t_i e^{-t_i \delta_g^{(i)}}.$$

Multiplying m such priors yields an aggregated pseudo-prior

$$P(\delta_g; \{t_i\}_{i=1}^m) = \exp\left(-\delta_g \sum_{i=1}^m t_i\right) \prod_{i=1}^m t_i \propto e^{-\delta_g \sum_{i=1}^m t_i}.$$

3.2 Half-life

Half-life is related to decay rate by

$$\lambda_g(\delta_g) = \frac{\log 2}{\delta_g}$$

with inverse $\delta_g(\lambda_g) = \frac{\log 2}{\lambda_g}$. GRAND-SLAM uses the plug-in estimate

$$\hat{\lambda}_{g,\text{plug-in}} = \frac{\log 2}{\hat{\delta}_g}.$$

This plug-in estimate is not the MAP estimate, even if the MAP estimate is used for $\hat{\delta}_g$.[5] By the change of variables formula, the posterior and log posterior distributions of λ_g are

$$p_{\Lambda_g}(\lambda_g) = d_g\left(\frac{\log 2}{\lambda_g}\right)\left|\frac{d\delta_g}{d\lambda_g}\right| = d_g\left(\frac{\log 2}{\lambda_g}\right)\frac{\log 2}{\lambda_g^2}$$
$$\log_{\Lambda_g}(\lambda_g) = \log d_g\left(\frac{\log 2}{\lambda_g}\right) - 2\log\lambda_g + \log(\log 2).$$

The MAP half-life estimate is therefore

$$\hat{\lambda}_{g,\text{MAP}} = \arg\max_\lambda \log_{\Lambda_g}(\lambda) = \arg\max_\lambda \log d_g\left(\frac{\log 2}{\lambda}\right) - 2\log\lambda$$

which can be found numerically.

Notably, it is provably distinct from the plug-in estimate. Specifically, the MAP half-life estimate must be a root of the first derivative of the log posterior:

$$\frac{d}{d\lambda}\log_{\Lambda_g}(\lambda) = \left[\frac{d}{d\delta}\log d_g(\delta)\Big|_{\delta=\frac{\log 2}{\lambda}} \cdot \frac{d\delta}{d\lambda}\right] - \frac{2}{\lambda} + 0.$$

Applying the plug-in estimate based on the decay rate MAP yields

$$\begin{aligned}\frac{d}{d\lambda}\log_{\Lambda_g}(\hat{\lambda}_{g,\text{plug-in}}) &= \left[\frac{d}{d\delta}\log d_g(\delta)\Big|_{\delta=\frac{\log 2}{\hat{\lambda}_{g,\text{plug-in}}}} \cdot \frac{d\delta}{d\lambda}\right] - \frac{2}{\hat{\lambda}_{g,\text{plug-in}}} \\ &= \left[\frac{d}{d\delta}\log d_g(\delta)\Big|_{\hat{\delta}_g} \cdot \frac{d\delta}{d\lambda}\right] - \frac{2}{\hat{\lambda}_{g,\text{plug-in}}} \\ &= -\frac{2}{\hat{\lambda}_{g,\text{plug-in}}}\end{aligned}$$

[5] In general, the MAP estimator is not functionally invariant, unlike the maximum likelihood estimator: "if $\hat{\theta}$ is the MLE for θ, and if $g(\theta)$ is any transformation of θ, then the MLE for $\alpha = g(\theta)$ is by definition $\hat{\alpha} = g(\hat{\theta})$." [Wikipedia]

where the last equality uses the fact that $\hat{\delta}_g$ is the MAP estimate for δ_g, so the first derivative of the log-posterior of δ_g must be 0 when evaluated at $\hat{\delta}_g$. Since the first derivative of the log-posterior of the half-life parameter λ_g is not 0 at the plug-in estimate, that estimate is not the mode of the posterior distribution.

3.3 Production rate

RNA production rate is related to decay rate by

$$\sigma_g = s(\delta_g) = \delta_g \cdot a_g$$

with inverse $\delta_g = s^{-1}(\sigma_g) = \frac{\sigma_g}{a_g}$, where a_g is the total (steady-state) RNA abundance. The plug-in estimator is simply

$$\hat{\sigma}_g = \hat{\delta}_g \cdot a_g.$$

Its interpretation depends on how one treats the total (steady-state) RNA abundance a_g:

- If a_g is considered a known, fixed constant, then the posterior distribution of σ_g is

$$p_\sigma(\sigma_g) = d(s^{-1}(\sigma_g)) \left| \frac{ds^{-1}}{d\sigma_g} \right| = d\left(\frac{\sigma_g}{a_g}\right) \frac{1}{a_g}$$

  Because linear transformations do not shift the mode, $\hat{\sigma}_g$ is a valid MAP estimator. Credible intervals are simply $[\delta_{\text{lower}} \cdot a_g, \delta_{\text{upper}} \cdot a_g]$.

- If a_g is instead considered a random variable containing biological and/or measurement noise, then one must consider the joint posterior $P(\sigma_g, \delta_g | \mathbf{y}, p_e, p_c)$.

TODO: (1) Explain how to utilize all timepoints/samples for estimating production rate. (2) Currently, the production rate is in dimensions of reads/time, which depends on sequencing depth and other auxilliary factors. Determine how to estimate a normalized production rate to allow comparing across sequencing depths and samples.